Computer Vision - HW Assignment 2

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Firstly, for the first task, the gaussian derivative kernel picks up edges of different sizes at different scales. The below examples illustrate this clearly. In the first image, the smaller kernel puts more emphasis on the texture of the sand, while the largest one tries to pick up larger patterns in it. In the second image, the largest kernel begins to show the larger edge of the bottle shading. A similar effect happens when keeping the kernel size fixed and scaling down the image, although the down-scaling may reduce details.

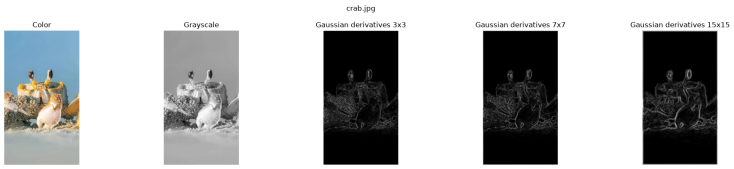


Figure 1: Gaussian Derivative kernel grayscale



Figure 2: Gaussian Derivative kernel grayscale



Figure 3: Boundaries detected using Gaussian derivative kernels on downscaled images.

Regarding the second task, the Generalized Boundaries performs better than the Generalized Derivative because it decouples the size of the kernel from the

"slope" of the edge model developed in the paper. That is to say, it may integrate information from a larger neighbourhood of the point while also weighing it appropriately, closer points being more important than more distant ones.

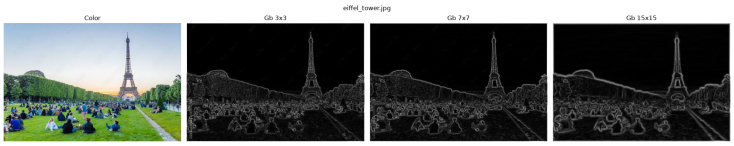


Figure 4: Generalized Boundaries

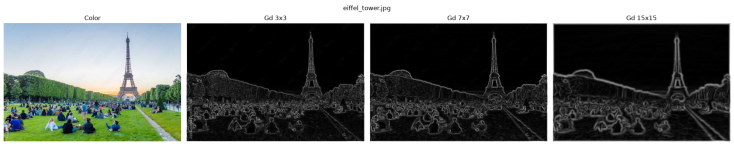


Figure 5: Generalized Derivatives